

anywhere between 34.9 years to 18.9 years depending on where in the cycle you start retirement.

TABLE 6.1 Anything Can Happen; Don't Leave It to Chance

Sequence of Returns	\$100 Initial Nest Egg and Spending			
	Market Cycle	\$6 Per Year	\$7 Per Year	\$8 Per Year
N.A.	Flat 5% Market	35.8 yrs.	25.1 yrs.	19.6 yrs.
+5,+10,+5,0	Retire into Bull	41.7	27.7	21.3
+5,0,+5,+10	Retire into Bear	31.4	22.7	18.0
+5,+15,+5,-5	Retire into Bull	50.1	31.0	23.4
+5,-5,+5,+15	Retire into Bear	27.9	20.8	16.5
+5,+20,+5,-10	Retire into Bull	65.6	34.9	25.5
+5,-10,+5,+20	Retire into Bear	24.9	18.9	15.4
+5,+25,+5,-15	Retire into Bull	Infinity	39.6	28.1
+5,-15,+5,+25	Retire into Bear	22.4	17.1	14.3

Source: M.Milevsky, "The Trigonometry of Retirement Income," *Research Magazine*, February 2007.

As you can see from the same table, if you invest in even more volatile asset classes that range from +25% to -15% per year, for example, the dispersion of outcomes is an even wider 17.1 years to 39.6 years.

Another important insight from the Table 6.1 is the impact of the spending rate itself. Notice that when you are spending \$8 per year, the gap between the “good” sequence and the “bad” sequence of returns is only $21.3 - 18 = 3.3$ years in the first rows (low risk) of the table. This same gap is $41.7 - 31.4 = 10.3$ years when spending is reduced to \$6 per year. At first glance this might seem odd. Why is the sequence-of-returns effect more powerful at lower spending rates? But, of course, this is a relative effect. If you spend more (that is, \$8) you will be worse off in terms of sustainability horizon regardless of the sequence, compared to only spending \$6 per year. That said, the gap between worst- and best-case scenario increases the less you spend. This impact becomes more pronounced at greater volatility levels. Withdrawing less might improve your sustainability odds somewhat, but it won't immunize you from a bad initial sequence of returns.